1. Clinical Question & Core Hypothesis

Clinical Question

What molecular mechanisms determine whether immune structures translate into clinical benefit?

Working Hypothesis

Primary hypothesis: In bladder cancer, a high stromal-to-tumor ratio (TSR) creates an extracellular-matrix–rich niche in which tumor cells expressing elevated GREM1 and MDK cooperate to remodel the ECM and invade lymphovascular spaces, increasing tumor cell dissemination and resulting in shorter progression-free intervals. Consequently, bladder tumors with high TSR and co-overexpression of GREM1 and MDK will have significantly shorter PFI compared with TSR-low or GREM1/MDK-low tumors, defining a stroma-driven, metastasis-prone subgroup that may benefit from anti-stromal or anti–MDK/GREM1 therapies. The hypothesis adheres to the constraint of a single imaging feature (TSR) and 1–2 anchor genes (GREM1, MDK).

Element	Value
Imaging feature	Stromal-to-tumor ratio (TSR)
Anchor gene(s)	GREM1, MDK
Primary endpoint	PFI

2. Why This Gene — Not Others

Primary hypothesis gene(s)

GREM1: GREM1) each forecast progression/recurrence, but it is unclear why some tumors with aggressive molecular signatures progress quickly while others do not — specifically, how tumor stromal architecture modulates the impact of pro-invasive molecules on dissemination and clinical progression.

MDK: MDK) and a molecular mediator (GREM1) each forecast progression/recurrence, but it is unclear why some tumors with aggressive molecular signatures progress quickly while others do not — specifically, how tumor stromal architecture modulates the impact of pro-invasive molecules on dissemination and clinical progression.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	CTCs	Supporting modifier from knowledge graph

Secondary	EV-mRNA signatures	Supporting modifier from knowledge graph
Secondary	KRT17	Supporting modifier from knowledge graph
Secondary	HDL-C	Supporting modifier from knowledge graph
Secondary	HRR	Supporting modifier from knowledge graph
Secondary	GLU	Supporting modifier from knowledge graph
Secondary	UA	Supporting modifier from knowledge graph
Secondary	microalbuminuria	Supporting modifier from knowledge graph
Secondary	SDC2	Supporting modifier from knowledge graph
Secondary	TFPI2	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: In cohorts with TSR and GREM1/MDK data, the interaction of TSR with GREM1/MDK expression will predict PFI beyond either factor alone.

Prediction 2: TSR-high tumors with elevated GREM1/MDK will show biomarkers of invasion/ECM remodeling and higher incidence of nodal or distant progression.

Prediction 3: In retrospective cohort data, the prognostic benefit of the imaging feature will be modified by the anchor gene expression level (significant interaction term).

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- CTCs: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- EV-mRNA signatures: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.